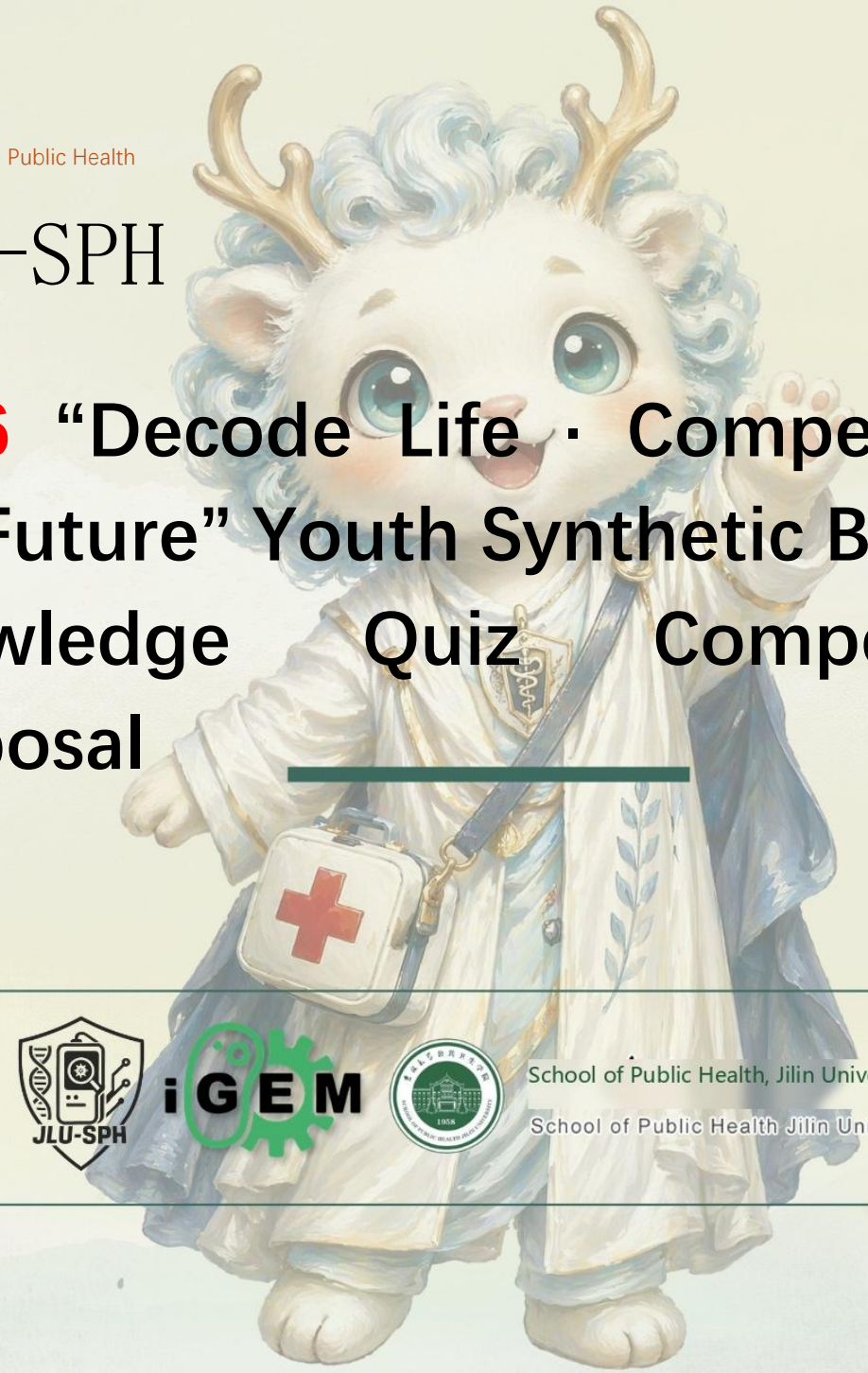
Protecting Life · Public Health

JLU-SPH

# 2026 “Decode Life · Compete for the Future” Youth Synthetic Biology Knowledge Quiz Competition Proposal



School of Public Health, Jilin University

School of Public Health Jilin University

JILIN UNIVERSITY · SCHOOL OF PUBLIC HEALTH

## Ⅰ、 Basic Information of the Activity

- Activity Title: "Decoding Life, Competing for the Future" Youth Synthetic Biology Knowledge Quiz Competition
- Activity Theme: Promoting learning through competition, unlocking the mysteries of synthetic biology , and lighting up the path of scientific exploration for young people
- Activity Type: ☑ Science Education & Popularization ☐ Public Welfare Service ☐ Cultural Promotion ☐ Community Service ☐ Other:
- Organizer: JLU-SPH Team

## Ⅱ、Activity Objectives

1. Core Objective: Utilizing knowledge competitions as a platform to disseminate fundamental concepts and the application value of synthetic biology among adolescents, inspiring interest in life sciences, dispelling the stereotype of synthetic biology as "inaccessible and difficult," and promoting the popularization of synthetic biology science.

2. Specific Goals:

○ Knowledge/Cognitive Aspect: Enable youth to understand fundamental knowledge points such as the definition of synthetic biology, its core technologies, and practical applications in daily life, as well as to clarify common cognitive misconceptions.

○ Behavioral/Skills Aspect: Enhance quick thinking, on-the-spot response, and teamwork capabilities, and foster habits of scientific information retrieval and summary.

○ Emotional/Attitudinal Aspect: Eliminate children's fear of pathogens and diagnostic equipment, stimulate their interest in life science exploration, and encourage habits such as frequent hand washing and proper mask wearing.

3. Expected Outcomes:

○ Number of Participants: Approximately 40-50 individuals.

○ Satisfaction Goal: Over 90%

○ Complete documentation of the entire activity process

## Ⅲ、Target Participants

- Core Group: Middle and high school students
- Participant Characteristics: Possess basic biological knowledge, strong curiosity, enjoy interactive competition formats, exhibit an eagerness to explore cutting-edge technology, suitable to absorb scientific knowledge through engaging quiz formats; they constitute a key group for deepening scientific interest and initiating career enlightenment.

## Ⅳ、 Date and Venue

- Date of the Activity: May 11, 2026
- Total Duration: Approximately 40 minutes
- Venue: School multimedia classroom

## Ⅴ、Activity Content and Procedure

[Overview of Activity Procedure]

| Stage | Time | Session Title | Specific Content | Responsible Personnel | Materials / Filming Focus |
|---|---|---|---|---|---|
| Stage | **40 min** | Pre-class Scientific Popularizati on | Team members conduct synthetic biology science communication in the class | Main lecturer, teaching assistants | Popular Science PPT |
| Stage | **20 min** | Venue and Equipment Preparation | Testing projection and audio equipment; preparing scoring tools, prizes, and questionnaires | General coordinator, logistics staff | PPT, scoreboard, cultural and creative prizes, questionnaire |
| Main Stage | **40 min** | Group Knowledge Quiz | Conducting three rounds of competition in groups: compulsory questions, buzzer questions, and risk questions | All teaching assistants, logistics staff | Tiered question bank, buzzer props |
| Main Stage | **10 min** | Review of Incorrect Questions + Open Q&A | Deconstructing and explaining frequently incorrect questions, opening the floor for students' queries and responses | Main lecturer | Handouts summarizing incorrect questions |
| Closing Stage | **10 min** | Activity Summary and Questionnaire Collection | Summary and award presentation, collection of paper-based satisfaction questionnaires | General coordinator, teaching assistants | Certificates, paper questionnaires |

**[Detailed Description of Each Stage]**

- **Stage 1: Preliminary Science Popularization Lecture**
- **Core Content:** Deconstructing the core concept of synthetic biology "designing biological components" , integrating popular science cases such as artificial proteins, microbiological remediation, self-developed portable respiratory tract detection devices, mRNA vaccines, and rapid home screening, to establish a comprehensive knowledge system and lay the foundation for the competition.
- **Key Message:** Synthetic biology is not limited to laboratory research; it can be practically implemented within households and schools, safeguarding the respiratory health of the elderly and children.
- **Stage 2: Group Knowledge Competition**

- **Core Content:** Students are randomly assigned to several groups and sequentially complete compulsory questions, buzzer questions, and risk questions. The competition topics cover curricular biology, fundamentals of synthetic biology, and knowledge related to civilian detection and epidemic prevention, balancing both basic and advanced content. Students are encouraged to share insights based on their own experiences with seasonal colds and home-based epidemic prevention, as well as to propose creative ideas for device optimization.
- **Key Points:** Priority will be given to works that are playful and effectively communicate epidemic prevention knowledge. Artistic proficiency is not the sole criterion for evaluation.

- **Stage 3: Review of Incorrect Answers & Open Q&A**
  - **Core Content:** Frequently missed questions from the competition are systematically reviewed and connected to the principles discussed in the lecture for detailed explanation. An open session is provided for free questioning.
  - Targeted responses are given to students' queries regarding various aspects of biotechnology and public health testing.
- **Stage 4: Award Ceremony & Conclusion**
  - **Core Content:** Certificates and science popularization cultural products are awarded to winning groups. The host summarizes the core of the activities, distributes satisfaction questionnaires, and all participants take a commemorative group photo.

## VI、 Personnel Configuration and Division of Labor

- **General Supervisor (1 person):** Responsible for overall coordination, liaising with the school, controlling workflows and timing, and handling emergencies. Must possess strong communication skills and the ability to oversee the entire event.
- **Main Lecturer (1 person):** Responsible for knowledge explanation, review of question banks, and in-depth analysis of incorrect answers. Must be familiar with synthetic biology knowledge and able to convey information in an engaging and accessible manner.
- **On-site Assistants (3 persons):** Responsible for distributing materials, maintaining order, scoring, and operating the buzzer system.
- **Photography & Logistics (2 persons):** Material transport, venue setup, and post-event cleaning.

## VII、 Materials List

- Equipment:
  - Laptop computer, projector
  - Camera/high-resolution smartphone
- Educational Materials:
  - Popular Science PPT
  - Competition question bank
  - Activity feedback questionnaires
- Logistics:

- Buzzer system
- Scoreboards
- Standard first aid kit
- Cleaning wipes, garbage bags

**Ⅷ、Risk Contingency Plan**

- Personnel Risk: Crowd congestion and minor injuries among students during the exhibition. Immediate evacuation of the crowd, use of the first aid kit for treatment, and simultaneous notification to the homeroom teacher.
- Venue Risk: Venue temporarily unavailable. Activate backup classroom and confirm alternative venues with school authorities in advance.
- Equipment Risk: Buzzer/projector malfunction. Switch to raising hands for answering, and use backup computers and whiteboards for supplementary explanations.

**IX. Effectiveness Evaluation**

- Process Evaluation: On-site observation, workflow documentation, and statistical analysis of correct answers.
- Satisfaction Evaluation: Collect paper-based feedback questionnaires and conduct random on-site oral interviews with students about their experiences. Gather responses immediately after the event concludes.